

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4304

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1. Online Symposium Portal

```
<form action="/register" method="POST">  
<input type="text" name="name">  
<input type="email" name="email">  
<button>Submit</button>  
</form>
```

1. Appropriate HTML5 Semantic Elements Missing

The structure lacks semantic elements such as:

- <header> – page heading/title
- <main> – main content area
- <section> – symposium registration section
- <label> – form field descriptions
- <footer> – footer information

1. Standards-Compliant Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>Symposium Registration</title>
</head>
<body>
<header>
  <h1>College Symposium Registration</h1>
</header>
<main>
  <section>
    <h2>Registration Form</h2>
    <form action="/register" method="POST">
      <label for="name">Name:</label>
      <input type="text" id="name" name="name" required>

      <label for="email">Email:</label>
      <input type="email" id="email" name="email" required>
      <button type="submit">Submit</button>
    </form>
  </section>
</main>
<footer>
  <p>&copy; 2026 College Symposium</p>
</footer>
</body>
</html>
```

2. HTTP Request Generated During Submission

POST /register HTTP/1.1

Host: www.college.edu

Content-Type: application/x-www-form-urlencoded

Interpretation

- POST → sends data to the server.
- /register → target resource.
- Host → server domain.
- Form data is placed in the request body.

3. HTTP Response Cycle

Step 1: Browser Sends Request

POST /register HTTP/1.1

Step 2: Server Processes Data

- Validates input.
- Stores registration details in database.

Step 3: Server Sends Response

HTTP/1.1 200 OK

Content-Type: text/html

<h1>Registration Successful</h1>

4. Usability and Accessibility Improvements

Usability

- Add placeholders.
- Use client-side validation.
- Display success/error messages.
- Group related fields using <fieldset>.

Accessibility

- Associate labels using for attribute.
- Ensure keyboard navigation.
- Provide ARIA attributes where necessary.
- Maintain sufficient color contrast.
- Use semantic HTML elements.

2. Cross-Browser Compatibility Issue

1. Improper Nesting and Missing Closing Tags

Problems:

- `<h1>` is not closed.
- `<p>` is not closed.
- Missing `<!DOCTYPE html>`.
- Generic `<div>` used instead of semantic elements

2. Browser interpretation

Different browsers may:

- Automatically insert missing closing tags.
 - Display the page differently.
 - Cause inconsistent layouts.
 - Produce unexpected rendering.
-

3. Missing semantic elements

Missing HTML5 semantic elements:

- `<header>`
- `<main>`
- `<section>`
- `<footer>`

4. Corrected W3C-compliant HTML

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Company</title>

</head>

<body>

<header>

  <h1>Welcome</h1>

</header>

<main>

  <section>

    <p>About us</p>

  </section>

</main>

<footer>

  <p>&copy; 2026 Company</p>

</footer>

</body>

</html>
```

5. Techniques for cross-browser compatibility

- Use valid HTML5 code.
- Validate using the W3C Validator.
- Test on multiple browsers.
- Use CSS reset or Normalize.css.
- Use responsive design.
- Use vendor prefixes where necessary.

- Avoid deprecated HTML tags.

3. Form Submission Failure

1. Issues in the form configuration

Problems:

- Button type is "button" instead of "submit".
 - Password is sent using the GET method.
 - GET is not secure for login credentials.
-

2. Why the HTTP request is not triggered

- A button with type="button" does not submit the form.
 - Only type="submit" triggers form submission automatically.
-

3. Role of GET method

- GET appends form data to the URL.
- Suitable for search forms.
- Not suitable for passwords because the data is visible in the URL and browser history.

4. Correct implementation

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Login Form</title>

</head>

<body>

<form action="/submit" method="POST">

  <label for="username">Username:</label>
```

```
<input type="text" id="username" name="username" required>
<label for="password">Password:</label>
<input type="password" id="password" name="password" required>
<button type="submit">Login</button>
</form>
</body>
</html>
```

5. Improvements for secure data transmission

- Use the **POST** method instead of GET.
- Enable **HTTPS** to encrypt data.
- Validate inputs on both client and server.
- Hash passwords before storing them.
- Use secure session management.
- Implement CSRF protection.
- Display clear error messages without exposing sensitive information.